

FLUIDS

Fluid: Any substance that does not have a definite shape

- Liquids –keep the same volume (incompressible)
- Gases –do not keep the same volume (are compressible)

Density

- Density = mass / volume

$$\rho = \frac{m}{V}$$

ρ = density (kg / m³)

m = mass (kg)

V = volume (m³)

- Density of water = 1000 kg / m³
- $\rho_{\text{object}} < \rho_{\text{water}}$ ☐ object floats
- $\rho_{\text{object}} > \rho_{\text{water}}$ ☐ object sinks

Some Common Densities:

Material	Density (kg/m ³)
Fresh water	1,000
Salt water	1,025
Iron	7,860
Mercury	13,600
Gold	19,300
Copper	8,890
Air	1.29

Buoyant Force & Archimedes' Principle

The Buoyant Force

- When an object floats, the upward buoyant force equals the downward pull of gravity.
- The buoyant force can float very heavy objects, and acts upon objects in the water whether they are floating, submerged, or even sitting on the bottom.

$$F_B = \rho_f V_f g$$

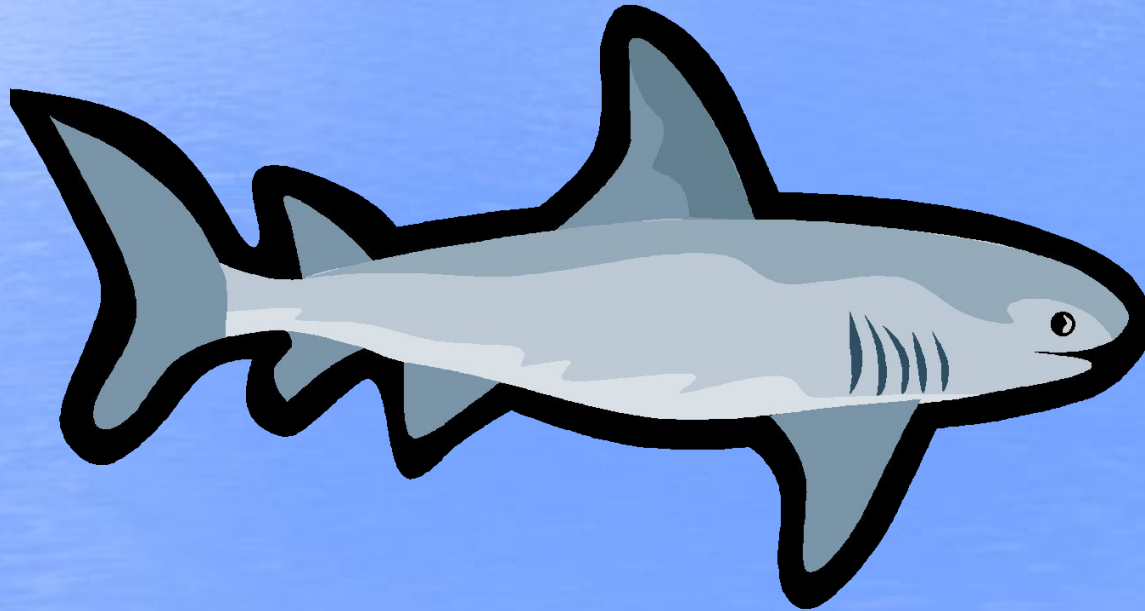
F_B : the buoyant force exerted on a submerged or partially submerged object.

V_f : the volume of displaced liquid.

ρ_f : the density of the displaced liquid.

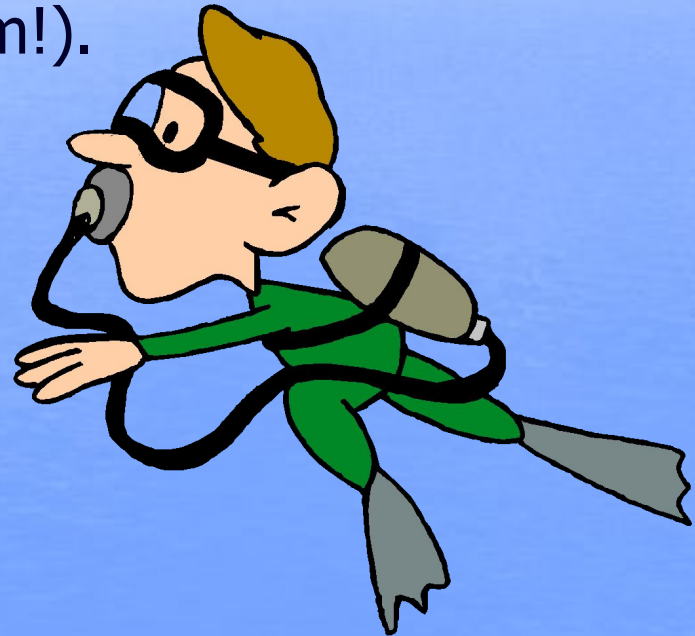
Buoyant force on submerged object

- A shark's body is not neutrally buoyant, and so a shark must swim continuously or he will sink deeper.



Buoyant force on submerged object

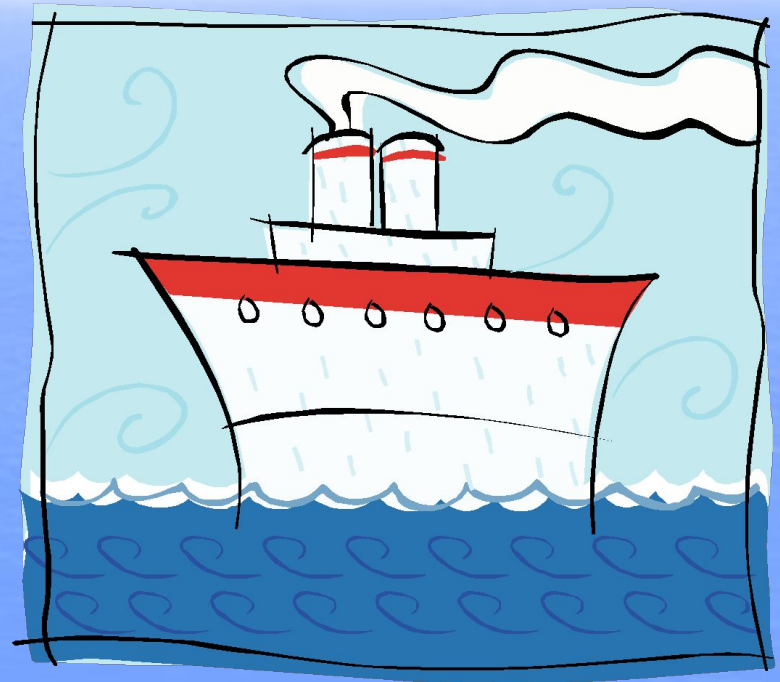
- SCUBA divers use a buoyancy control system to maintain neutral buoyancy (equilibrium!).



- If the diver wants to rise, he inflates his vest, which increases the amount of water he displaces, and he accelerates upward.

Buoyant force on floating object

- If the object floats on the surface, we know for a fact $F_{\text{buoy}} = mg$! The volume of displaced water equals the volume of the submerged portion of the ship.



Sample problem – Bouyant Force

- What is the buoyant force on a spherical helium balloon floating in air if it has a radius of 50 cm? The density of air is 1.29 kg/m^3 .

Buoyant Force

- The buoyant force can be extremely strong.
- Incredibly massive objects can float, even when they are not intended to...

Parking in St. Bernard Parish after Hurricane Katrina







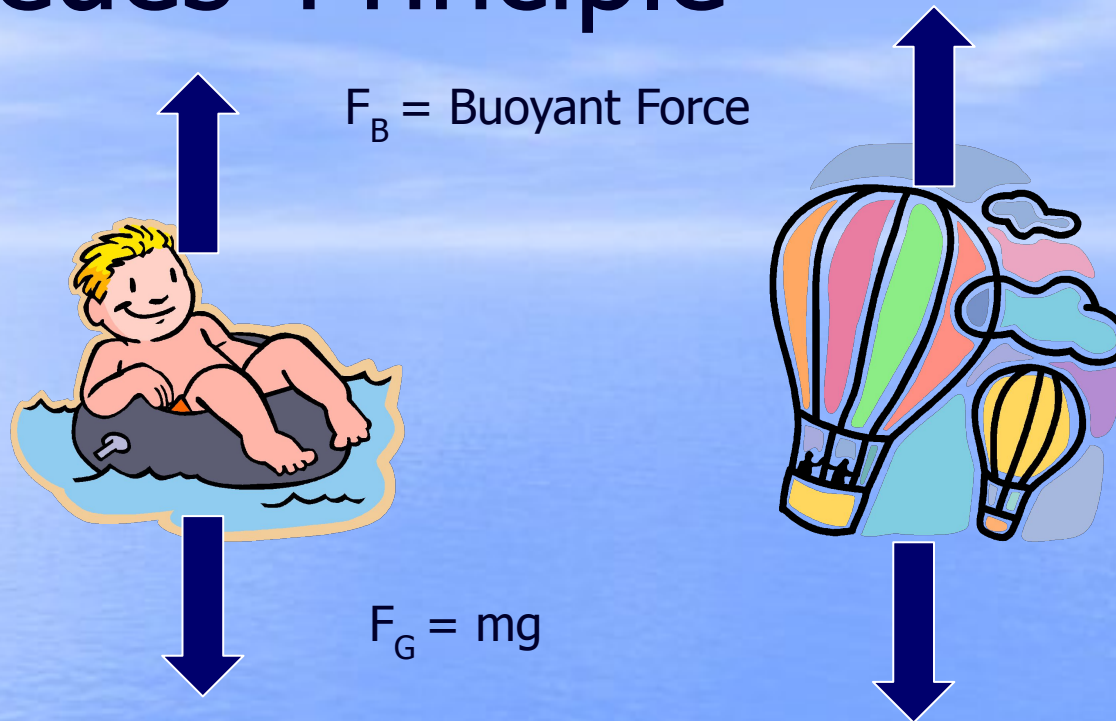
Weight

- Density = mass / volume
- Weight is different than density!

$$W = mg = (\rho V)g$$

- Why do we care about the *weight* of a fluid?.....

Archimedes' Principle



- Floating is a type of equilibrium
- An upward force counteracts the force of gravity for these objects. This upward force is called the buoyant force.
- Archimedes' Principle: a body immersed in a fluid is buoyed up by a force that is equal to the weight of the fluid it displaces.

Archimedes' Principle



- Archimedes' Principle: A body immersed in a fluid displaces an amount of water equal to its own weight...OR
- A body immersed in a fluid is buoyed up by a force (F_B) which is equal to the weight of the displaced fluid.

$$F_B = W_{fluid}$$

$$\therefore F_B = \rho_f V_f g$$

F_B = Buoyant Force (N)

ρ_f = density of displaced fluid
(kg / m³)

V_f = volume of displaced fluid

g = 9.8 m / s²

Archimedes' Principle



- If an object FLOATS:
 - $F_B = mg$
- If an object SINKS (submerged):
 - $W_A = W - F_B$

W_A = Apparent Weight ("underwater weight")

W = Actual Weight (on dry land)

F_B = Buoyant Force

Sample problem – Apparent Weight

- A cannon built in 1868 in Russia could fire a cannonball with a mass of 480 kg. When suspended from a scale and submerged in fresh water ($\rho_{\text{water}} = 1 \text{ E } 3 \text{ kg / m}^3$), a cannonball of this type has an apparent weight of $4.07 \text{ E } 3 \text{ N}$. What is the volume of the cannonball?

Water Bridge in Germany... What a feat!

This is a channel-bridge over the River Elbe and joins the former East and West Germany, as part of the unification project. It is located in the city of Magdeburg, near Berlin.





Question:

Did the bridge have to be designed to withstand the additional weight of ship and barge traffic, or just the weight of the water?



Answer:

It only needs to be designed to withstand the weight of the water!

Why?



- A ship always displaces an amount of water that weighs the same as the ship, regardless of how heavily a ship may be loaded.
- This is how concrete and steel ships float!

Density & Buoyant Force – Sample Problem

- What is the volume of a 5 kg block of metal with a density of $3.57 \times 10^3 \text{ kg / m}^3$? How large is the buoyant force when the block is submerged in water ($\rho_{\text{water}} = 1 \times 10^3 \text{ kg / m}^3$? What is the apparent weight of the block when it is submerged in water? (0.0014 m^3 ; 13.73 N ; 35.27 N)

Density & Buoyant Force – Sample Problem

- What is the apparent weight of a helium ($\rho_{\text{He}} = .79 \text{ kg} / \text{m}^3$) balloon filled to a volume of $.127 \text{ m}^3$ as it floats upward through the air ($\rho_{\text{air}} = 1.29 \text{ kg} / \text{m}^3$)?

Fluids FRQ

- Now work 2005B5! 10 minutes....
- We'll check and grade "AP-Style"

Fluids THQ

- HW: Problems # 1-9 on the THQ

Fluid Pressure

Pressure

$$P = \frac{F}{A}$$

P : pressure (Pascals)

F: force (N)

A: area (m²)



- Pressure unit:
 - Pascal (1 Pa = 1 N/m²)
- *The force on a surface caused by pressure is always normal (or perpendicular) to the surface. This means that the pressure of a fluid is exerted in all directions, and is perpendicular to the surface at every location.*

Sample Problem – Force & Pressure

- 1) Calculate the net force on an airplane window if cabin pressure is 90% of the pressure at sea level, and the external pressure is only 50% of that at sea level. Assume the window is 0.43 m tall and 0.30 m wide.

Pascal's Law

- Pascal's Law – A change in pressure at one point in an enclosed fluid is distributed equally to all points of the fluid.
- Ex: Toothpaste – will come out of the tube no matter where you squeeze it!



- Pascal's Law – A change in pressure at one point in an enclosed fluid is distributed equally to all points of the fluid.

Hydraulic Press

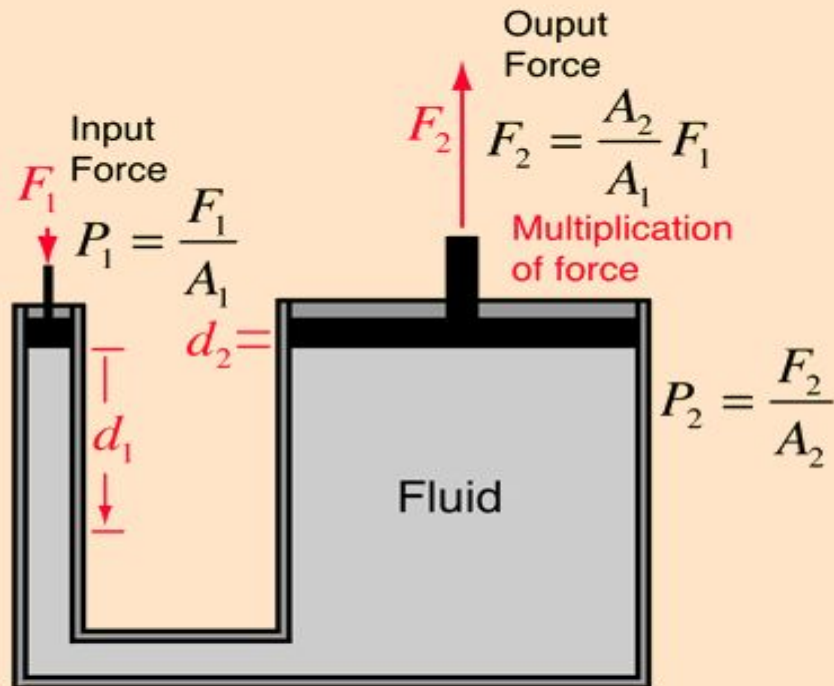
A multiplication of force can be achieved by the application of fluid pressure according to Pascal's principle, which for the two pistons implies

$$P_1 = P_2$$

This allows the lifting of a heavy load with a small force, as in an auto hydraulic lift, but of course there can be no multiplication of work, so in an ideal case with no frictional loss:

$$W_{\text{input}} = W_{\text{output}}$$

Calculation



$$F_1 d_1 = F_2 d_2$$

$$d_1 = \frac{F_2}{F_1} d_2 = \frac{A_2}{A_1} d_2$$

You have to pay for the multiplied output force by exerting the smaller input force through a larger distance.

Pascal's Law

- Pascal's Law – A change in pressure at one point in an enclosed fluid is distributed equally to all points of the fluid.

Pascal's Principle

Pressure is transmitted undiminished in an enclosed static fluid.

$A_1 = 5 \text{ cm}^2$



$A_2 = 500 \text{ cm}^2$

$F_1 = 10 \text{ N}$ Applied force to the stopper

$P_1 = \frac{10 \text{ N}}{5 \text{ cm}^2} = 2 \text{ N/cm}^2$

Pressure is transmitted undiminished in an enclosed static fluid.

$F_2 = P_2 A_2 = (2 \text{ N/cm}^2)(500 \text{ cm}^2)$
 $= 1000 \text{ N!!}$ plus the force from the weight of the liquid.

Resulting force on bottom of jug.

$P_2 = P_1 + \rho gh$

Static fluid pressure

Like a liquid lever, changing areas in an enclosed fluid permit multiplication of force

Any externally applied pressure is transmitted to all parts of the enclosed fluid, making possible a large multiplication of force ([hydraulic press principle](#)). The pressure at the bottom of the jug is equal to the externally applied pressure on the top of the fluid plus the [static fluid pressure](#) from the weight of the liquid.

Pressure & Hydraulics

- Pressure has applications to hydraulic systems (like the brakes in your car)

Automobile Hydraulic Lift

A hydraulic lift for automobiles is an example of a force multiplied by [hydraulic press](#), based on [Pascal's principle](#). The fluid in the small cylinder must be moved much further than the distance the car is lifted.

Pressure is exerted on fluid in small cylinder, usually by a compressor.



Pressure is exerted equally in all parts of an enclosed static fluid: Pascal's law



Though the pressure is the same, it is exerted over a much larger area, giving a multiplication of force that lifts the car.

The force in the small cylinder must be exerted over a much larger distance. A small force exerted over a large distance is traded for a large force over a small distance.

Pressure & Hydraulics

- Pressure has applications to hydraulic systems (like the brakes in your car)

For example, if the lift cylinder were 25 cm in diameter and the small cylinder were 1.25 cm in diameter, then the ratio of the areas is 400, so the hydraulic press arrangement gives a multiplication of 400 times the force. To lift a 6000 newton car, you would have to exert only $6000 \text{ N} / 400 = 15 \text{ N}$ on the fluid in the small cylinder to lift the car. However, to lift the car 10 cm, you would have to move the oil $400 \times 10 \text{ cm} = 40 \text{ meters}$. This is practical by pumping oil into this small cylinder with a small compressor.

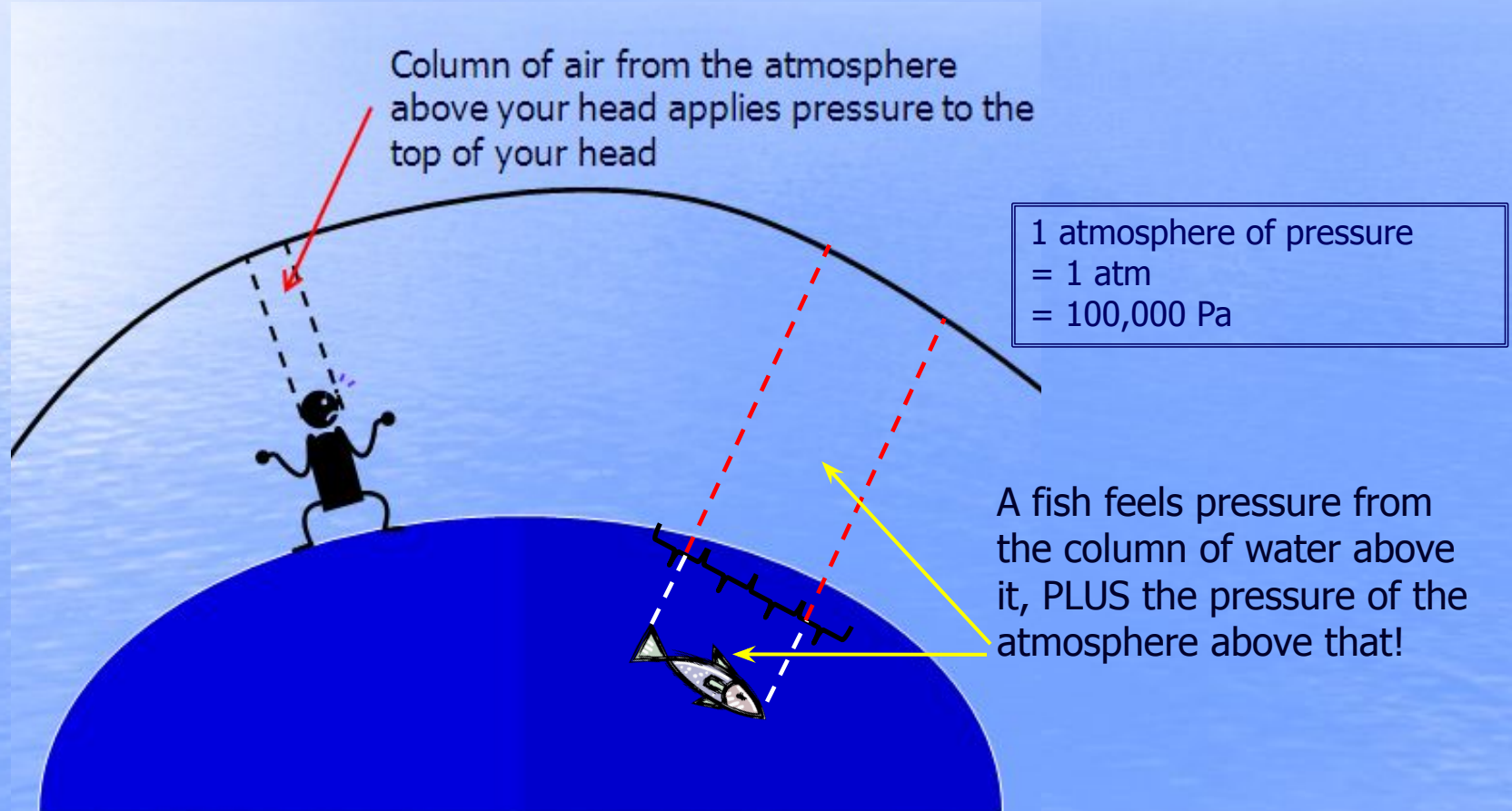
Sample Problem – Pascal's Law

- **SAMPLE PROBLEM:** You apply 5 N of force to the brake pedal in you car, which drives a piston with a radius of 0.5 cm into a cylinder. If the weight of your car is 9800 N, how large of an area do your brake pads cover?

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

The Pressure of a Fluid

- A column of fluid (liquid or gas) above an object puts Pressure on that object



The Pressure of a Fluid

$$P = \rho gh$$

P: pressure (Pa)

ρ : density (kg/m³)

g: 9.8 m/s²

h: height of liquid column (m)
above the object

- This type of pressure is often called *gauge pressure*.
- *Absolute pressure* is obtained by adding the atmospheric pressure to the hydrostatic (water) pressure.

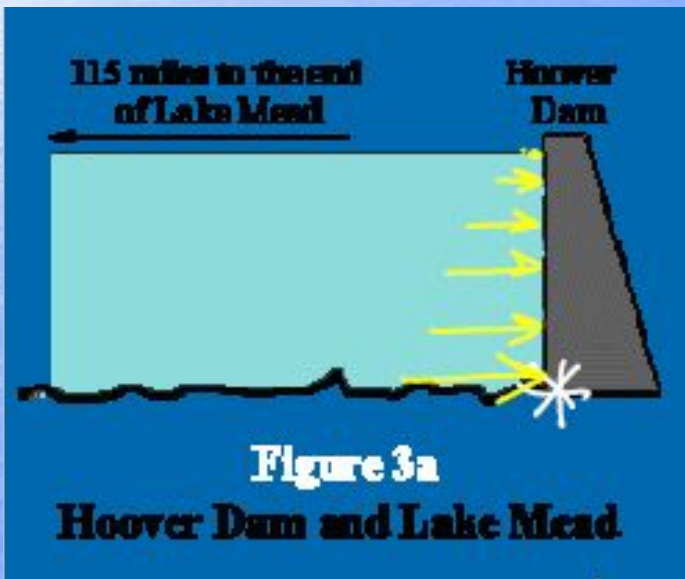
$$P_{abs} = P_{atm} + \rho gh$$

P_{abs} : Absolute (Total) pressure (Pa)

$P_{atm} = 100,000 \text{ Pa}$ (= 1 atmosphere)

Sample Problem – Pressure at a Depth

- The depth of Lake Mead at the Hoover Dam is 600ft. What is the gauge (or hydrostatic) pressure and what is the absolute pressure at the base of the dam?



No such thing....

- In Physics, there are no forces that “suck”
- In other words, there is no such thing as a “suck force”
- STRAWS work on differences in pressure:
 - You decrease the pressure in your mouth
 - The atmospheric pressure on the surface of the liquid is then higher than in your mouth
 - The liquid is *pushed* up the straw
- Physics doesn't “suck....”
- ...it “blows.” 😊



Fluids FRQ

- Now work 2003B6 (10 min),
- We'll check and grade "AP-Style"
- 2004B2 save for HW/Review (15 min)

Fluids THQ

- HW: Problems # 10-15 on the THQ

Fluid Flow Continuity

When a Fluid Flows...

- ...mass is conserved.
- Provided there are no inlets or outlets in a stream of flowing fluid, the same mass per unit time must flow everywhere in the stream.
- ***The volume per unit time of a liquid flowing in a pipe is constant throughout the pipe.***
- We can say this because liquids are not compressible, so mass conservation is also volume conservation for a liquid.

Fluid Flow Continuity

$$V = Avt$$

V: volume of fluid (m³)
A: cross sectional area
at a point in the pipe
(m²)
v: speed of fluid flow at
a point in the pipe
(m/s)
t: time (s)



Dividing V
by t yields:

$$Q = Av$$

Q: volume flow rate
(m³/s) (=Vol/time)
A: cross sectional area
at a point in the pipe
(m²)
v: speed of fluid flow at
a point in the pipe
(m/s)

Fluid Flow Continuity

$$A_1 v_1 = A_2 v_2$$

A_1, A_2 : cross sectional
areas at points 1 and 2

v_1, v_2 : speed of fluid
flow at points 1 and 2

- The smaller the cross-sectional area, the faster the fluid flows (thumb over a garden hose)

Sample Problem – Fluid Flow

- A pipe of diameter 6.0 cm has fluid flowing through it at 1.6 m/s. How fast is the fluid flowing in an area of the pipe in which the diameter is 3.0 cm? How much water per second flows through the pipe?

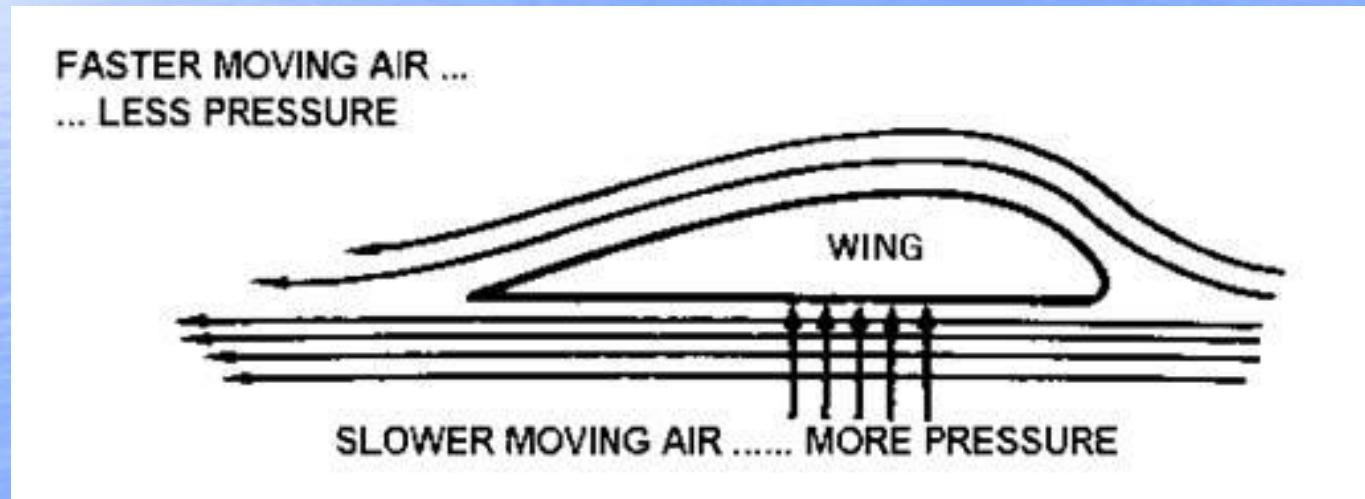
Sample Problem – Fluid Flow

- The water in a canal flows 0.10 m/s where the canal is 12 meters deep and 10 meters across. If the depth of the canal is reduced to 6.5 meters at an area where the canal narrows to 5.0 meters, how fast will the water be moving through this narrower region?

Bernoulli's Theorem

Bernoulli's Principle

- **Bernoulli's Principle – The faster a fluid flows, the lower the pressure**
 - Airplane wings – “Lift”



Bernoulli's Theorem

Pressure

"Kinetic Energy"

"Potential Energy"

$$P + \rho gy + \frac{1}{2} \rho v^2 = \text{const.}$$

P : pressure (Pa)

ρ : density of fluid (kg/m³)

g: 9.8 m/s²

h: height above lowest point (m)

v: speed of fluid flow at a point in the pipe (m/s)

- All other considerations being equal, when fluid moves faster, the pressure drops.

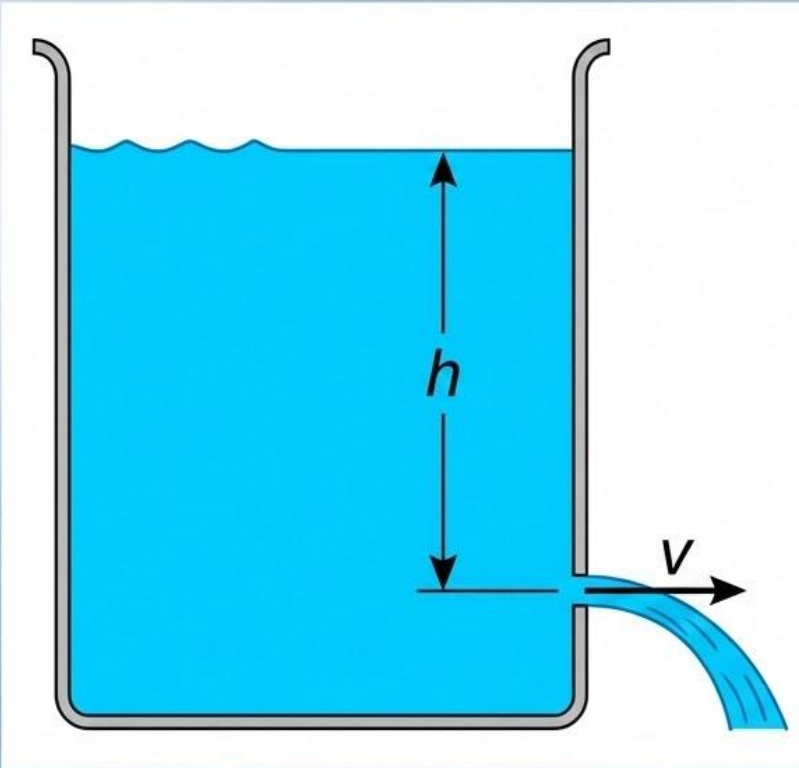
Bernoulli's Principle and Moving Fluids

- The faster a fluid flows, the lower the pressure
- In a hurricane or tornado, the high winds traveling across the roof of a building can actually lift the roof off the building.
- Fabric convertible tops “flap” in the wind
- Nasty hotel shower curtains stick to you, blech!

Sample Problem – Bernoulli's Principle

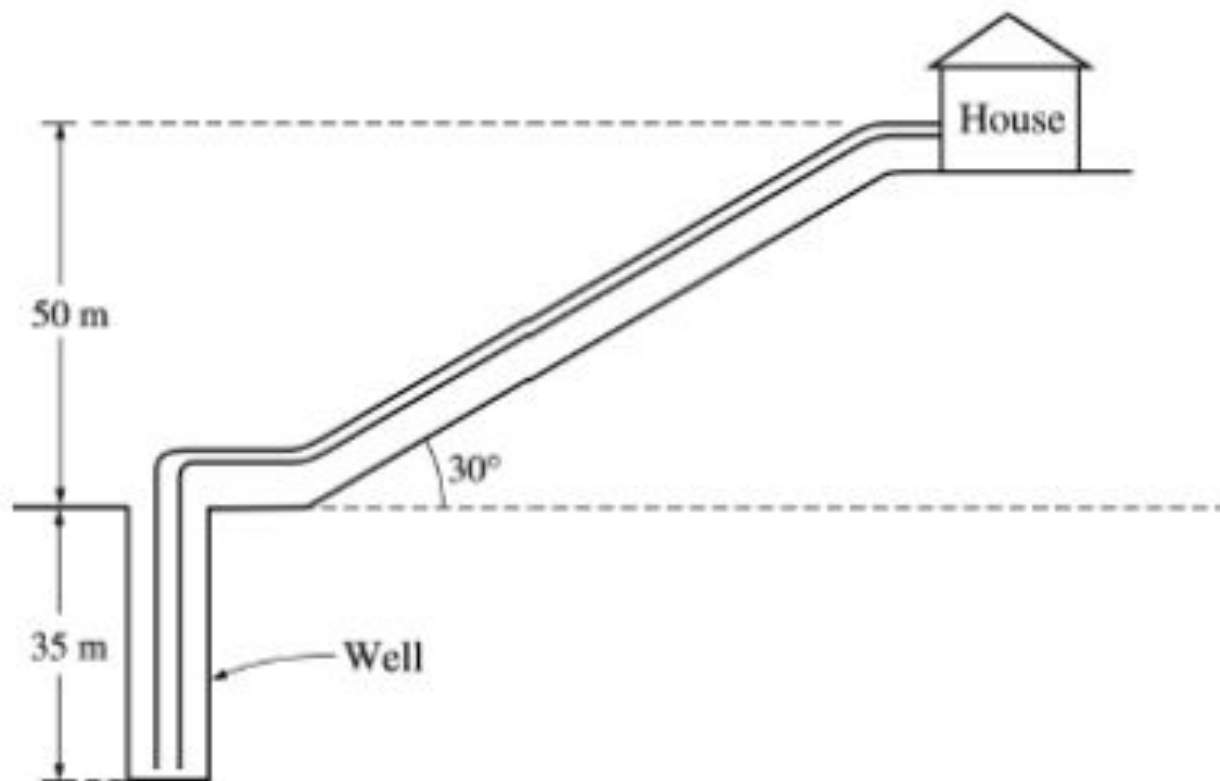
- Because of an aneurism, the cross-sectional area (A_1) of the aorta increases to a value of $A_2 = 1.7A_1$. The speed of the blood (density = 1060 kg/m^3) through a normal portion of the aorta is $v_1 = 0.40 \text{ m/s}$. Assuming the aorta is horizontal, by how much does the pressure in the abnormal region exceed the pressure in the normal region?

Torricelli's Law



$$v = \sqrt{2gh}$$

Torricelli's theorem states that the speed of efflux, v , of a fluid through a sharp-edged hole at the bottom of a tank filled to a depth h is the same as the speed that a body (in this case, a drop of water) would acquire in falling freely from a height h , i.e., $v = \sqrt{2gh}$, where g is the acceleration due to gravity.



- (b) In the well, the water flows at 0.50 m/s and the pipe has a diameter of 3.0 cm . At the house the diameter of the pipe is 1.25 cm .
- Calculate the flow velocity at the house when a faucet in the house is open.
 - Calculate the pressure at the well when the faucet in the house is open.

Sample Problem – Bernoulli's Principle

- FRQ 2009B3 (let's work it together!)

Fluids Lab Challenge

- **Challenge:** A cup is placed on the edge of the lab table and filled to a given height h with water (density = 1000 kg/m^3). A hole has been punched in the side of the bottle near the bottom to allow water to flow out of the hole. It is your job to place a “catcher cup” where you think the water will land, if it is allowed to spray over the edge of the table.
 - Develop a theory indicating what measurements you will take and how you will determine where the water will land. Be sure to indicate what equations you will use and how you will determine each variable. A sketch will be helpful.
 - Collect all equipment you will need, and run the lab according to your theory. You must show ALL work to receive full credit. Indicate clearly the distance that you calculated.
- If your water lands in the catch cup you will receive an automatic 100 for a quiz grade! If it does not, you will need to turn in your theory and calculations to be graded!

Fluids THQ

- HW: Problems # 16 - end on the THQ